

Tirth Patel

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Education

Western Michigan University

Aug 2022 – Dec 2024

MS in Computer Science

- **Coursework:** Design and Analysis of Algorithms, Computer Network, Computer Systems, Information Retrieval, DBMS, IoT Security, Adv. Storage Retrieval and Processing of Big Data, Adv. AI, Data Science.

Gujarat Technological University

Aug 2017 – Jul 2021

BE in Computer Engineering

- **Coursework:** Data Structures, Theory of Computation, System Programming, Software Engineering, DBMS, Web Data Management, Object-Oriented Programming Principles, Advanced JAVA, Mobile Computing, Wireless Comm., AI, Data Mining-Business Intelligence.

Skill

Programming: Advanced in C/C++, Python, Java, JavaScript, TypeScript, Assembly, Swift, HTML5, CSS.

Databases: MySQL, MongoDB, DynamoDB, PostgreSQL.

Libraries and Tools: Flask, Bootstrap, Django, Sass, ReactJS, NodeJS; Protocols: TCP/IP, BGP, Nmap, Wireshark, Kafka, Data analysis tools (Pandas, NumPy), AI modeling tools (TensorFlow, PyTorch).

Technologies: React, .NET, WordPress, Docker, Kubernetes, Wireshark, Excel, Linux, iOS/macOS Development, AWS, Cybersecurity protocols, CI/CD Code pipelines.

Experience

Software Engineer & Strategist

Kalamazoo, MI

WMU Dept. of Management & Office of Sustainability

Apr 2024 – Dec 2024

- Designed, developed, and deployed a full-stack Bronco Challenge for Sustainable Impact Application & an interest-matching platform using:
 - Frontend technologies: JavaScript, HTML5, CSS3, TypeScript, and UI/UX design principles (e.g., responsive design, accessibility standards).
 - Backend solutions: Firebase, Vercel, and WMU-CMS.
 - Development tools and methodologies: Git (version control), Agile frameworks, SDLC.
- Enhanced user engagement and participant retention by creating a multi-page, interactive, and user-friendly web application, driving a 2.5X increase in User participation rates.
- Directed end-to-end software development & marketing initiatives, collaborating with stakeholders, cross-functional teams to meet objectives. Streamlined team communication workflows to align project deliverables with goals and deadlines.

Software Engineer

Ahmedabad, India

TZO Technologies

Mar 2020 – Jun 2022

- Engineered and deployed high-performance web applications using JavaScript frameworks (React, Node.js), RESTful APIs, and MongoDB within AWS cloud infrastructure.
- Utilized Agile software development methodologies, implementing unit testing, integration testing, and end-to-end testing to ensure system reliability, functionality, and scalability.
- Acquired proficiency in multiple programming languages and technologies, including C#, C++, React JS, JavaScript, JSON. Managed continuous integration and delivery processes to streamline development cycles, ensuring efficient and scalable deployments.
- Leveraged Git for version control and facilitated collaborative development through rigorous code reviews, testing, and debugging to maintain code quality standards.

Teaching Assistant - Computer Engineering

Ahmedabad, India

Alpha College of Engineering & Technology

Jun 2018 – Feb 2020

- Supported and taught sections of Data Structures & Algorithms, Computer Networks, Object-Oriented Programming, C, Java, Web Technology.
- Supervised and led teaching assistant teams to maintain consistent and high-quality instruction. Took on primary teaching responsibilities during faculty absences, delivering lectures and facilitating discussions.
- Designed and implemented comprehensive course summaries and study guides, improving student learning outcomes and comprehension. Consistently received top-tier evaluations from students, reflecting strong pedagogical skills and a commitment to academic excellence.

Projects

Urban Tree Management – Responsive Web Application [🔗](#)

- Designed and developed a GIS-enabled web application for assigning unique tree IDs to improve identification and access to plantation data, achieving exceptional precision in location datasets.
- Spearheaded the development of unique tracking software using technologies including React, JavaScript, HTML, CSS, Bootstrap, and Google Maps API, significantly enhancing system efficiency.
- (Publication) "Urban Tree Management-UTM", Guided by Professor Ajaykumar T. Shah. International Journal of Science and Research Technology, 7 Issue 3, (March 2021) ISSN (Online): 2395-1052. [🔗](#)

Information Retrieval System – PyTerrier

- Explored and implemented advanced indexing techniques, including tokenization, stemming, and stopword removal, to enhance data retrieval accuracy.
- Engineered three distinct weighting algorithms (PL2, BM25, TF-IDF) to enable document retrieval, indexing, ranking, and evaluation.

IoT Security – Traffic Analysis & Vulnerability Assessment

- Utilized Lehmer mean and time-based windowing techniques to analyze network traffic and detect anomalies, categorizing them into attack and benign datasets.
- Applied Wireshark for capturing and examining network traffic, leading to the identification of critical patterns and vulnerabilities with actionable insights.

Adaptive Smart Traffic Light Control System

- Simulated an adaptive traffic control system in Python, integrating fixed-time, traffic-responsive, and multi-phase strategies for traffic optimization.
- Applied libraries like NumPy, Pandas, and Matplotlib for data analysis, visualization, and interface development.
- Implemented AI/ML-driven traffic prediction and optimization algorithms using time series analysis and reinforcement learning, enhancing system efficiency and reducing congestion.

CogniDrive: Real-Time Adaptive Perception for Autonomous Driving

- Designed an autonomous driving system utilizing Convolutional Neural Networks (CNNs) for real-time object detection, tracking, and behavior prediction.
- Implemented Reinforcement Learning (RL) for decision-making and navigation in dynamic environments.
- Developed a Conditional Imitation Learning (CIL) model, integrating sensory data with high-level commands, achieving robust performance in urban driving scenarios through noise injection and data augmentation.

Friend Recommendation System for Multiplayer Games Using AWS

- Architected a serverless friend recommendation system leveraging AWS services (Neptune, Lambda, API Gateway, Cognito) and Node.js.
- Developed a scalable data model in Amazon Neptune and implemented backend logic for generating personalized recommendations, optimizing user engagement.

UAV-CAN Anomaly Detection

- Engineered a Python-based framework to detect anomalies in UAVCAN communication data using data science techniques like generalized mean and R-value analysis.
- Conducted data preprocessing and feature engineering (e.g., time windowing) to enhance model accuracy, achieving high anomaly detection rates.
- Enhanced UAV cybersecurity by designing solutions to detect and mitigate communication-based cyberattacks, gaining hands-on experience in data science and system resilience.

Certification

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| ◦ Web Development by University of Michigan | ◦ Google Agile Project Management |
| ◦ Advance Python Course | ◦ DevOps - Amazon Web Services |
| ◦ Programming with JavaScript | ◦ iOS App Development with Swift |

Leadership & Extracurricular activities

- Led a team in Hackathon at Alpha Anveshan TechFest, hosted by Gujarat Technological University, Ahmedabad.
- Gained insights into Adani Mundra Port's on-network system and supply chain software through industrial visit.
- Amongst Top 3 Contestants in Nirma University's inter-university coding competition.
- Engaged member of Western Michigan University's Google Developer Student Club, Blockchain Crypto Club.